

Notes: Mullainathan and Shleifer (AER, 2005)

The Market for News

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1 Big picture

- **Question.** Does competition among newspapers eliminate media bias, or can competitive media markets still produce slanted news?
- **Setting.** Newspapers observe information about an underlying state of the world and choose how to report it. Readers may be rational information seekers, or they may prefer news that confirms their prior beliefs.
- **Core challenge.** The standard economic intuition says that competition should discipline news providers: if one outlet slants, another outlet can report the truth and attract readers. The paper shows that this logic depends crucially on what readers want.
- **Main conceptual contribution.** The paper builds a demand-side theory of media bias. Bias does not need to come from owners, journalists, regulators, or parties. It can arise because profit-maximizing newspapers cater to readers who enjoy confirmatory news.
- **Main modeling innovation.** Newspapers choose a *slanting strategy*. Readers have beliefs in a one-dimensional belief space. Competition then resembles a Hotelling model: newspapers choose positions in belief space, and readers prefer nearby positions.
- **Benchmark result.** If readers are rational and care only about accuracy, newspapers do not slant. Monopoly already delivers unbiased reporting; duopoly only lowers prices.
- **Homogeneous-belief result.** If all readers share the same biased belief, newspapers slant toward that common belief. Competition does not improve accuracy; it simply drives price down.
- **Heterogeneous-belief result.** If readers disagree, duopolists segment the market and slant in opposite directions. Individual newspapers become more extreme, and the average reader may encounter more biased news.
- **Accuracy result.** Heterogeneity can nevertheless improve the information available to a *conscientious reader* who reads multiple sources. Opposing slants omit opposite facts, so cross-checking different outlets can reveal a more complete picture.
- **Main takeaway.** Reader heterogeneity is more important for aggregate media accuracy than competition per se. Competition without reader diversity can reinforce common bias; competition with reader diversity can create polarized outlets whose aggregate information is more balanced.

2 Data and measurement (key objects)

2.1 Why this is a theory paper

- This is not an empirical paper. There is no dataset, no natural experiment, and no estimated regression coefficient.
- The paper instead provides a formal theory of how reader preferences and market structure shape media slant.

- The “measurement” objects are therefore the model’s primitives and outcome variables:
 - the underlying state being reported,
 - the data observed by newspapers,
 - reader beliefs,
 - reader preferences over slant and belief confirmation,
 - newspaper slanting strategies,
 - newspaper prices,
 - average reader bias,
 - and conscientious reader bias.

Interpretation. The paper is best read as a clean benchmark model. It asks what a competitive media market would do even without ideological owners or partisan journalists.

2.2 Underlying state and newspaper signal

Readers care about an underlying variable t , such as the state of the economy or the truth about a political scandal:

$$t \sim N(0, v_t). \quad (1)$$

Newspapers observe a noisy data realization:

$$d = t + \varepsilon, \quad \varepsilon \sim N(0, v_\varepsilon). \quad (2)$$

A newspaper reports:

$$n = d + s, \quad (3)$$

where s is the newspaper’s slant.

Interpretation. The paper separates the raw facts d from the way those facts are presented n . Slant is the gap between reported news and underlying data.

2.3 Reader beliefs

Readers may hold a biased prior belief about the expected value of the state. A reader with belief b expects news closer to b .

The paper studies two belief environments.

- **Homogeneous beliefs.** All readers share the same belief b . Examples include broadly shared stereotypes or common national beliefs about foreign countries.
- **Heterogeneous beliefs.** Reader beliefs are distributed uniformly over an interval $[b_1, b_2]$, with mean $\bar{b}$. Examples include politically divisive issues where left- and right-leaning readers expect different narratives.

Interpretation. The distinction between common beliefs and dispersed beliefs is the central comparative static in the paper. It determines whether competition produces common slant or polarized slant.

2.4 Reader utility: rational versus biased demand

A rational reader wants accurate information and dislikes slant because slant makes it harder to infer the truth:

$$U^r = \bar{u} - \chi s^2 - P. \quad (4)$$

A biased reader dislikes slant, but also dislikes news that is far from her belief b :

$$U^b = \bar{u} - \chi s^2 - \phi(n - b)^2 - P. \quad (5)$$

Since $n = d + s$, biased-reader utility can also be written as:

$$U^b = \bar{u} - \chi s^2 - \phi(d + s - b)^2 - P. \quad (6)$$

- $\chi > 0$ measures the cost or disutility of slant itself.
- $\phi > 0$ measures the strength of demand for confirmatory news.
- P is the newspaper price.

Interpretation. Readers are not modeled as loving lies. They dislike excessive slant. But they also enjoy news that fits their beliefs. Newspapers trade off these two forces.

2.5 Newspaper strategy and timing

The newspaper chooses a slanting rule $s(d)$ and a price P .

1. Newspaper(s) announce slanting strategies.
2. Newspaper(s) announce prices.
3. Readers decide whether to buy.
4. Newspapers observe data d and report $n = d + s(d)$.
5. Readers consume the news and receive utility.

The paper studies monopoly and duopoly.

- **Monopoly.** One newspaper chooses a slanting strategy and price.
- **Duopoly.** Two newspapers observe the same data and choose slanting strategies and prices.

Interpretation. Competition matters because it changes both prices and product positioning in belief space.

2.6 Average reader bias

For homogeneous readers, average reader bias is:

$$\text{ARB}_{\text{hom}} = \mathbb{E}_d[(n - d)^2]. \quad (7)$$

For heterogeneous readers, where reader i reads news n_i , average reader bias is:

$$\text{ARB}_{\text{het}} = \int_i \mathbb{E}_d[(n_i - d)^2] di. \quad (8)$$

Interpretation. Average reader bias asks how far the news consumed by the typical reader is from the data observed by the newspaper.

2.7 Conscientious reader bias

The paper also studies a hypothetical conscientious reader who reads all available outlets and cross-checks them.

If the conscientious reader effectively observes n_c , then:

$$\text{CRB} = \mathbb{E}_d[(n_c - d)^2]. \quad (9)$$

Interpretation. Average reader bias and conscientious reader bias can move in opposite directions. Competition can increase bias for the average single-source reader while reducing bias for someone who compares opposing sources.

3 Models and empirical specifications (all equations + interpretation)

3.1 The optimal slanting rule toward a belief point

A convenient object used throughout the paper is the slanting rule toward a belief point B :

$$s_B(d) \equiv \frac{\phi}{\chi + \phi}(B - d). \quad (10)$$

This implies reported news:

$$n = d + s_B(d) = \frac{\phi}{\chi + \phi}B + \frac{\chi}{\chi + \phi}d. \quad (11)$$

Interpretation. Reported news is a weighted average of the data d and the target belief B . When demand for confirmation ϕ is high, reporting moves closer to B . When the cost of slanting χ is high, reporting stays closer to the data.

3.2 Benchmark: rational readers

If readers are rational, utility is:

$$U^r = \bar{u} - \chi s^2 - P. \quad (12)$$

Because slant only lowers utility, the newspaper chooses:

$$s_{\text{hom}}^* = s_{\text{het}}^* = 0. \quad (13)$$

For a monopolist:

$$P_{\text{hom}}^* = P_{\text{het}}^* = \bar{u}. \quad (14)$$

For duopolists:

$$s_{j,\text{hom}}^* = s_{j,\text{het}}^* = 0, \quad P_{j,\text{hom}}^* = P_{j,\text{het}}^* = 0. \quad (15)$$

Interpretation. With rational information-seeking readers, competition is not needed to obtain accuracy. Even a monopolist reports without slant because slant reduces readers' willingness to pay. Competition affects price, not truthfulness.

3.3 Homogeneous biased readers: monopoly

When all readers hold the same belief b , a monopolist chooses:

$$s_{\text{hom}}^*(d) = \frac{\phi}{\chi + \phi}(b - d). \quad (16)$$

The price is:

$$P_{\text{hom}}^* = \bar{u} - \frac{\chi\phi}{\chi + \phi}(b^2 + v_d), \quad (17)$$

provided expected utility is high enough for readers to buy.

The resulting report is:

$$n = \frac{\phi}{\chi + \phi}b + \frac{\chi}{\chi + \phi}d. \quad (18)$$

Interpretation. A monopolist slants toward the audience's common belief because that raises willingness to pay. The newspaper does not fully ignore the data because readers still dislike excessive slant.

3.4 Comparative statics of homogeneous-reader slant

The magnitude of slant rises with the strength of confirmatory preferences:

$$\frac{\partial |s_{\text{hom}}^*(d)|}{\partial \phi} > 0. \quad (19)$$

It falls with the cost of slanting:

$$\frac{\partial |s_{\text{hom}}^*(d)|}{\partial \chi} < 0. \quad (20)$$

Interpretation. Media bias is strongest when readers intensely demand confirmatory stories and when slanting is cheap.

3.5 Homogeneous biased readers: duopoly

With two newspapers and homogeneous biased readers, both newspapers choose the same slant:

$$s_{j,\text{hom}}^*(d) = \frac{\phi}{\chi + \phi}(b - d), \quad j = 1, 2. \quad (21)$$

Competition drives price to zero:

$$P_{j,\text{hom}}^* = 0. \quad (22)$$

Interpretation. Competition does not reduce slant when all readers share the same bias. Both firms must cater to the same confirmatory demand. They compete away price, not bias.

3.6 Heterogeneous biased readers: monopoly

Suppose reader beliefs are uniformly distributed on $[b_1, b_2]$ and the mean belief is $\bar{b} = 0$. If the belief dispersion is small enough, a monopolist covers the whole market and chooses:

$$s_{\text{het}}^*(d) = s_{\bar{b}}(d) = \frac{\phi}{\chi + \phi}(\bar{b} - d) = -\frac{\phi}{\chi + \phi}d. \quad (23)$$

The price is:

$$P_{\text{het}}^* = \bar{u} - \frac{\phi\chi}{\chi + \phi}v_d - \phi b_2^2. \quad (24)$$

If belief dispersion is too large, the monopolist does not cover the entire market.

Interpretation. A monopolist facing a diverse audience chooses a relatively moderate slant. When the audience is balanced around zero, the monopolist slants toward the average belief. This is analogous to locating near the middle of a Hotelling market.

3.7 Heterogeneous biased readers: duopoly and polarization

Now suppose two newspapers choose linear slanting strategies and beliefs are distributed symmetrically around zero, so $b_1 = -b_2$.

For sufficiently small belief dispersion, the duopoly equilibrium has:

$$s_{1,\text{het}}^*(d) = \frac{\phi}{\chi + \phi} \left(\frac{3}{2}b_1 - d \right), \quad (25)$$

$$s_{2,\text{het}}^*(d) = \frac{\phi}{\chi + \phi} \left(\frac{3}{2}b_2 - d \right), \quad (26)$$

and prices:

$$P_{j,\text{het}}^* = \frac{6\phi^2}{\chi + \phi} b_2^2, \quad j = 1, 2. \quad (27)$$

The reported news by newspaper j is:

$$n_j = d + s_{j,\text{het}}^*(d) = \frac{\phi}{\chi + \phi} \frac{3}{2}b_j + \frac{\chi}{\chi + \phi} d. \quad (28)$$

Interpretation. Duopolists locate outside the range of reader beliefs. Each newspaper slants toward a point more extreme than its most extreme readers. This softens price competition by increasing differentiation.

3.8 Why duopoly becomes more extreme

The logic is the same as spatial differentiation.

- If two newspapers choose similar slants, they compete for the same readers and prices are low.
- By choosing very different slants, each newspaper gains a more captive audience.
- The price benefit from differentiation can dominate the market-share loss from moving away from moderate readers.
- The equilibrium therefore involves polarization, not convergence to the median belief.

Interpretation. Competition can create incentives to become more extreme because differentiation is a way to avoid direct price competition.

3.9 Average reader bias under competition

When readers are heterogeneous and the monopoly covers the market, competition increases average reader bias:

$$\text{ARB}_{\text{mon,het}}(v_d) < \text{ARB}_{\text{duo,het}}(v_d). \quad (29)$$

Interpretation. More competition can make the news consumed by the average single-source reader more biased. The reason is market segmentation: each outlet caters to a narrower and more extreme audience.

3.10 Slanting as selective omission of facts

To study cross-checking, the paper gives a concrete technology of slant. A newspaper observes a string of facts f_i :

$$\Pr(f_i) = \begin{cases} +1 & \text{with probability } qg(d), \\ -1 & \text{with probability } q(1 - g(d)), \\ A & \text{with probability } 1 - q. \end{cases} \quad (30)$$

Here $+1$ is a positive fact, -1 is a negative fact, and A is no fact. A non-slanted newspaper reports the whole string. As the string becomes long, the ratio of positive to reported facts recovers the data:

$$\frac{N^+(f)}{N^-(f) + N^+(f)} \rightarrow g(d). \quad (31)$$

A newspaper slants upward by omitting negative facts. It slants downward by omitting positive facts.

Interpretation. The paper does not require newspapers to fabricate information. Bias can arise through selective emphasis and selective omission.

3.11 Cross-checking and conscientious readers

Let s_1 and s_2 be the slants of two newspapers. The conscientious reader's cross-checking function is:

$$x_c(s_1, s_2) = \begin{cases} \min\{s_1, s_2\}, & s_1 > 0, s_2 > 0, \\ \max\{s_1, s_2\}, & s_1 < 0, s_2 < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

The effective news exposure is:

$$n_c = \begin{cases} n, & \text{if there is one newspaper,} \\ d + x_c(s_1, s_2), & \text{if there are two newspapers.} \end{cases} \quad (33)$$

Conscientious reader bias is:

$$\text{CRB} = \mathbb{E}_d[(n_c - d)^2]. \quad (34)$$

Interpretation. If both newspapers slant in the same direction, cross-checking only partially helps. If they slant in opposite directions, each reveals facts the other omits, and the conscientious reader can recover an unbiased picture.

3.12 The bottom-line accuracy result

With heterogeneous readers, duopoly lowers conscientious reader bias relative to monopoly:

$$\text{CRB}_{\text{het,duo}} < \text{CRB}_{\text{het,mon}}. \quad (35)$$

Under duopoly, heterogeneous readers also generate lower conscientious reader bias than homogeneous readers:

$$\text{CRB}_{\text{het,duo}} < \text{CRB}_{\text{hom,duo}}. \quad (36)$$

Interpretation. Polarization is bad for the average reader who consumes only one source, but it can be good for a reader who compares sources. The paper’s central message is not that bias disappears, but that diverse bias can be easier to aggregate than common bias.

4 Identification and instruments (what makes the theoretical results credible)

4.1 No empirical identification or instruments

- The paper does not estimate a causal effect from data.
- It does not use an instrumental variable, difference-in-differences design, randomized experiment, or structural estimation.
- Its credibility comes from the clarity of the model and from the robustness of the mechanism across market structures.

Interpretation. For this paper, the relevant question is not “what identifies the coefficient?” but “which assumptions generate the equilibrium result?”

4.2 Key identifying assumptions in the model

The core conclusions depend on three assumptions.

1. **Readers value confirmation.** Biased readers prefer reports that are closer to their prior beliefs.
2. **Newspapers can slant.** They can present the same data in different ways, especially by selecting or omitting details.
3. **Newspapers maximize profits.** They choose slant and price to attract readers and extract surplus.

Interpretation. These assumptions convert reader psychology into market-level media bias.

4.3 What competition can and cannot discipline

- Competition disciplines price strongly: duopoly drives prices down in the homogeneous-reader case.
- Competition does not discipline slant when readers want confirmation.
- With homogeneous readers, both newspapers converge to the same biased reporting strategy.
- With heterogeneous readers, competition induces differentiation and can make individual outlets more extreme.

Interpretation. The model separates price competition from quality competition. Competitive pressure lowers prices, but if readers define quality as confirmatory interpretation, competition can reduce accuracy.

4.4 Demand-side bias versus supply-side bias

The paper intentionally abstracts from supply-side ideological motives.

- Supply-side bias would come from owners, editors, journalists, or regulators wanting to push an ideology.
- Demand-side bias comes from readers rewarding newspapers for confirming their beliefs.
- The model shows that supply-side ideology is not necessary for media bias.

Interpretation. Even neutral, profit-maximizing newspapers can produce biased coverage if their audiences prefer biased coverage.

4.5 Why reader heterogeneity is central

Reader heterogeneity changes the market in two ways.

- It creates demand for differentiated outlets.
- It creates the possibility that opposing outlets reveal different pieces of information.

This is why the paper says heterogeneity is more important for accuracy than competition itself.

Interpretation. Competition is valuable only when there are different audiences for firms to serve. Without heterogeneous readers, competition duplicates the same bias.

4.6 Limits of the model

- The model is stylized and one-dimensional. Real news has many dimensions: ideology, tone, topic choice, source selection, expertise, and factual uncertainty.
- The paper treats readers' beliefs as given, though it notes that political entrepreneurs and media themselves may help create or reinforce those beliefs.

- The conscientious reader is a useful benchmark, but many real readers do not consume and integrate multiple sources.
- The cross-checking result depends on the idea that opposing outlets omit different facts rather than all distorting the same facts in the same way.
- The paper does not directly test whether actual media outlets behave according to the model.

Interpretation. The model is not a complete account of media markets. It is a mechanism: when readers enjoy confirmation, competition can amplify slant.

4.7 Relation to broader economic ideas

The paper contributes to a broad literature on whether rational, profit-maximizing firms eliminate the effects of biased consumers.

- The answer here is no: firms may cater to biases rather than eliminate them.
- The logic is similar to models in finance, politics, and consumer markets where sophisticated firms profit from biased agents.
- The paper's distinctive point is that competitive differentiation can make persuasion more extreme.

Interpretation. Market competition does not automatically debias consumers. Sometimes competition makes firms better at serving, reinforcing, and segmenting those biases.

5 Tables and figures

This paper has no conventional empirical tables or figures. The key exhibits are the formal propositions and the comparison of equilibrium cases. The following tables summarize the model's main objects and results.

5.1 Exhibit 1: Core model objects

Table 1: Main objects in the model

Object	Definition	Economic meaning
t	Underlying state	The truth or economic/political fact that readers care about.
$d = t + \varepsilon$	Newspaper data	The raw information observed by media outlets.
s	Slant	The distortion between data and reported news.
$n = d + s$	Reported news	What readers actually see.
b	Reader belief	The prior view the reader likes to see confirmed.
χ	Cost of slant	How much readers dislike excessive manipulation or incomplete reporting.
ϕ	Confirmation preference	How much readers dislike news far from their beliefs.
P	Newspaper price	The monetary price of reading the paper.

Read. The model separates facts, reporting, beliefs, and prices.

Economic message. Bias is a market outcome: newspapers choose s in response to reader preferences and competition.

5.2 Exhibit 2: Equilibrium slant by reader type and market structure

Table 2: Equilibrium slant across cases

Readers	Market structure	Equilibrium slant	Main implication
Rational	Monopoly	$s^* = 0$	Accuracy obtains without competition.
Rational	Duopoly	$s_1^* = s_2^* = 0$	Competition lowers price, not bias.
Homogeneous biased	Monopoly	$s^*(d) = \frac{\phi}{\chi + \phi}(b - d)$	Monopoly slants toward shared reader belief.
Homogeneous biased	Duopoly	$s_j^*(d) = \frac{\phi}{\chi + \phi}(b - d)$	Competition does not reduce common slant.
Heterogeneous biased	Monopoly	$s^*(d) = \frac{\phi}{\chi + \phi}(\bar{b} - d)$ if market is covered	Monopoly chooses a moderate slant toward average belief.
Heterogeneous biased	Duopoly	$s_j^*(d) = \frac{\phi}{\chi + \phi}(\frac{3}{2}b_j - d)$	Duopolists polarize and locate beyond the most extreme readers.

Read. The equilibrium depends less on the number of newspapers than on the combination of reader psychology and belief dispersion.

Economic message. Competition eliminates bias only in the rational-reader benchmark. With biased readers, competition either reproduces common slant or creates polarized slant.

5.3 Exhibit 3: Competition versus heterogeneity

Table 3: What improves accuracy?

Scenario	Effect on average single-source reader	Effect on conscientious reader
More competition with homogeneous readers	Little or no improvement in accuracy; price falls.	Little improvement because outlets carry similar slant.
More competition with heterogeneous readers	May increase individual-source bias through polarization.	Can improve accuracy because different outlets reveal different omitted facts.
More reader diversity	Creates demand for multiple perspectives.	Improves cross-checking value when all sources are accessible.
Common reader bias	Encourages common slant.	Hard to correct because all outlets omit similar facts.

Read. The effect of competition is ambiguous unless we know the distribution of reader beliefs and how readers consume news.

Economic message. Diversity of viewpoints is the key input into aggregate accuracy.

5.4 Exhibit 4: Cross-checking logic

Table 4: How a conscientious reader uses multiple outlets

Slant pattern	Facts omitted	Cross-checking result
Both outlets slant upward	Both tend to omit negative facts.	Cross-checking only partially reduces bias.
Both outlets slant downward	Both tend to omit positive facts.	Cross-checking only partially reduces bias.
One outlet slants upward and one downward	Each outlet omits facts on the opposite side.	Cross-checking can recover the full set of facts.
Only one outlet exists	No independent source to compare.	The reader is exposed to the outlet's slant.

Read. Opposite slants are especially valuable to a reader who checks multiple sources.

Economic message. Polarization is not purely harmful in the model. It harms one-source readers but can help multi-source readers reconstruct the truth.

6 Short summary

- The paper asks whether media competition eliminates bias.
- If readers are rational and want accuracy, newspapers do not slant; competition only lowers price.
- If readers want belief-confirming news, newspapers slant toward readers' priors.
- With homogeneous biased readers, competition does not reduce bias because both firms cater to the same shared belief.
- With heterogeneous biased readers, competition segments the market and makes individual outlets more extreme.
- This can increase the bias faced by the average single-source reader.
- But heterogeneous competition can reduce bias for a conscientious reader who compares outlets, because opposing slants reveal different omitted facts.
- The core result is that reader heterogeneity, not competition alone, is the main force that can improve aggregate media accuracy.

7 Conclusion

7.1 Core conclusion

Mullainathan and Shleifer show that competitive media markets do not automatically produce accurate news. If readers prefer stories that confirm their beliefs, profit-maximizing newspapers may slant toward those beliefs. Competition can lower prices, but it need not lower bias.

7.2 Economic intuition

The model is a Hotelling-style theory of persuasion. Reader beliefs are locations in product space. Newspapers choose locations by selecting slants. When readers are homogeneous, all firms choose the same slant. When readers are heterogeneous, firms differentiate and may become more extreme than their readers.

7.3 Main takeaway

The market for news rewards accuracy only when readers demand accuracy. If readers demand confirmation, media firms supply confirmation. The best hope for aggregate accuracy is not competition by itself, but competition combined with diverse reader beliefs and access to multiple independent sources.